EXPERIMENT: 08

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Subject Name:- System Design

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1. Aim:-

To design a high-level architecture for a scalable stock trading platform (similar to Zerodha, Groww, and Upstox) that supports real-time stock trading, order management, and portfolio tracking with high availability and low latency.

2. Objectives:-

The objectives of this experiment are:

1. To understand the architecture of a real-time stock trading system.
2. To design APIs for user authentication, KYC verification, market data, and order execution.
3. To analyze functional and non-functional requirements of a trading platform.
4. To study how real-time data streaming and event-driven systems work.
5. To ensure consistency in financial transactions while maintaining scalability.
6. To understand how portfolio and trade management systems operate.

3. Procedure:-

• Studied real-world stock trading platforms such as Zerodha, Groww, and Upstox to understand system behavior.
• Identified key system entities including Users, Stocks, Orders, Trades, Portfolio, and Watchlist.
• Analyzed the complete workflow from user registration → KYC → viewing stocks → placing orders → trade execution.
• Gathered functional requirements such as real-time stock updates, order placement, and portfolio tracking.
• Analyzed non-functional requirements including low latency, high availability, fault tolerance, and scalability.
• Designed RESTful APIs for user management, market data, portfolio, and watchlist features.
• Studied WebSocket-based real-time updates for stock prices.
• Understood event-driven architecture using message queues (Kafka) for trade execution and notifications.
• Evaluated trade-offs between strong consistency (orders, trades) and eventual consistency (analytics, notifications).

4. Functional Requirements:-

Users should be able to register, log in, and complete KYC verification.
• Users should be able to view real-time stock prices and historical data.
• The system should support placing, modifying, and canceling orders.
• Orders should provide accurate real-time status updates.
• Users should be able to maintain a watchlist with live updates.
• The system should provide a dashboard to track holdings, trade history, and P&L.
• Portfolio management should reflect executed trades instantly.

5. Non Functional Requirements:-

• The system must support millions of users simultaneously.
• Real-time stock updates should have very low latency (<100 ms).
• High availability (99.9%+) must be ensured for trading operations.
• Strong consistency is required for order execution and trades.
• The system should be fault tolerant and resilient to failures.
• Scalability should be achieved using microservices architecture.
• Secure communication and authentication (JWT, HTTPS) must be implemented.

6. Core Entities:-

• User
• Stock
• Order
• Trade
• Portfolio
• Watchlist

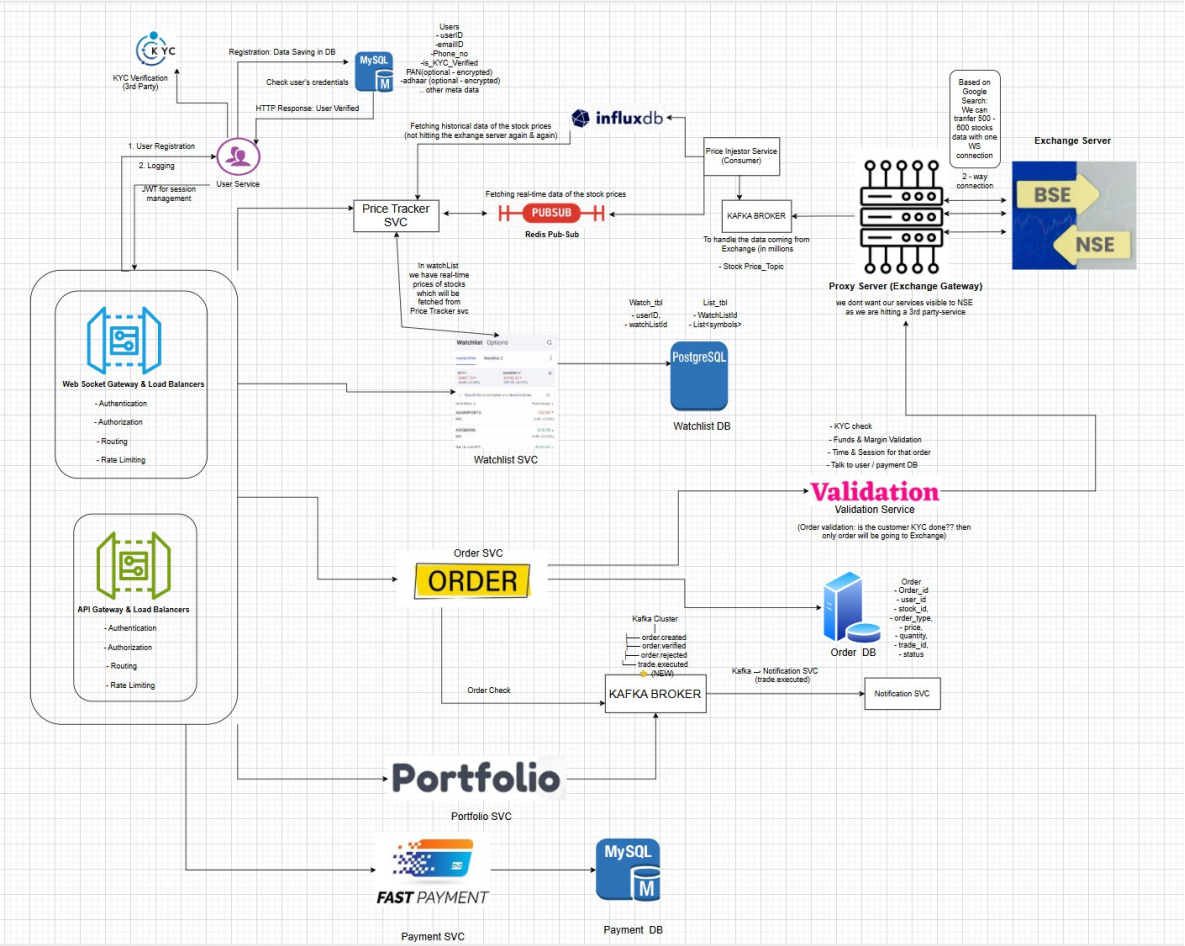
7. API Design:-

User APIs:
• POST /auth/signup
• POST /auth/login
• POST /auth/verify-KYC
Market Data APIs:
• WS /api/v1/stocks → Real-time stock updates
• GET /api/v1/stocks → Get all stocks
• GET /api/v1/stocks/{stock_symbol_id} → Stock details
• GET /api/v1/stocks/{stock_symbol_id}/history → Historical data
Order APIs:
• POST /api/v1/orders → Place order
• PUT /api/v1/orders/{order_id} → Modify order
• DELETE /api/v1/orders/{order_id} → Cancel order
• GET /api/v1/orders → Get user orders

Portfolio APIs:

- GET /api/v1/portfolio
- GET /api/v1/portfolio/holdings
- GET /api/v1/portfolio/positions
- GET /api/v1/portfolio/performance (Optional)

Watchlist APIs:

- GET /api/v1/watchlists
- POST /api/v1/watchlists/{watchlist_id}/symbols
- DELETE /api/v1/watchlists/{watchlist_id}/symbols/{symbol}

8. HLD:-